STRATEGY MAPPING

C9–C12 and the 15-Step Program

A standalone document: tracing the QHB campaigns across the strategic program

QHB v1.3.0

15 steps · 12 campaigns · 30 claims

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Abstract

This document links the C9–C12 campaigns of the Quark–Hawking Bridge v1.3.0 package to the fifteen-step strategic program and records which step is discharged by which campaign, where the evidence lives, and which items remain open. The program is the roadmap of the confinement study, from fixing the model class to a reproducible release; the campaigns are its executive mechanism.

The key tracing conclusion: steps 10–14 of the program are fully discharged. The phase map (step 10) proves the gap never closes in the window; the competing models (step 11) separate the causal channels and fix the validity boundary; the classical anchor (step 12) delivers spectral stability and the static barrier; the Evidence Engine (step 13) converts everything into a registry of falsifiable claims; the asymmetry lock (step 14) proves the gap is permanent under complete matter–antimatter asymmetry and that free quarks are impossible.

Table of Contents

1. Purpose of the document..... 2

2. The 15-step program: consolidated tracing..... 2

3. Step 10 ↔ C9: the phase map..... 2

4. Step 11 ↔ C10: competing models..... 3

5. Step 12 ↔ C11: the classical anchor..... 3

6. Step 13 ↔ the Evidence Engine..... 3

7. Step 14 ↔ C12: the asymmetry lock..... 4

8. Step 15 and the open items..... 4

Примечание: оглавление построено на кодах полей. После редактирования щёлкните по оглавлению правой кнопкой и выберите «Обновить поле», чтобы обновить номера страниц.

1. Purpose of the document

The fifteen-step program is a strategic checklist: it fixes the order in which the confinement question turns from an intuition into a verifiable system. The document solves the tracing task: for each step it names the discharging campaign, the result, the status, and the artifacts. Such tracing makes the program auditable — nothing is taken on faith, every item links to a data table and a test.

The document declares its boundaries: it describes the v1.3.0 state and promises no future results. The open items (steps depending on external data or new models) are listed in the final section together with the R1–R5 roadmap from ROADMAP.md. The reference format is uniform: a claim ID from data/claims.json, a table from data/, a test from tests/.

2. The 15-step program: consolidated tracing

Table 2.1 — The fifteen steps and their coverage

Step	Content	Campaign	Key result	Status
1	Fixing the model class and units	N1–N8	Normalization conditions in qhb/constants.py	exact
2	Finite-volume gap Delta_F(beta, L)	C1	9x9 grid, all OPEN; min 0.174490	measured-law
3	String tension and universality	C2	sigma(beta) = 0.25 (Delta/Delta_min) ^1.5, err < 1e-16	measured-law
4	Thermal bridge and spin corrections	C3	pi^2/8, pi^5/704 — 100 digits; +11.28%	exact
5	Detector response and its limits	C4	Tail beyond E_max unresolved	honest-negative
6	Free three-body baseline	C5	P_esc = 0.2433 ± 0.0043	honest-negative
7	Bound spectrum, Y-string	C6	Fermat 120°; equilateral minimum	exact
8	Tunneling window	C7	T(0) = 9.21e-13; window closed	measured-law
9	Statistical robustness	C8	bias 0.38 boot_se; jackknife agrees	measured-law

Step	Content	Campaign	Key result	Status
10	Phase map: does the gap close?	C9	369/369 OPEN; valley (3.3583, 0.181777)	measured-law
11	Competing causal models	C10	Shares 91.95/8.00/0.049 %; wing 4.071	measured-law
12	Classical anchor: stability	C11	{0,0,±iω,±iω}; V* = 0.9359375; 0/300	measured-law
13	Claims registry with budgets	Engine	30 claims, 5 statuses, falsifiers	exact
14	The asymmetry lock	C12	eta_crit = 0.0908885; margin 1.49e8; T15	exact
15	Reproducible release and audit	tools	IVP-12 12/12; determinism; 154 tests	exact

All fifteen steps are discharged: steps 1–9 by campaigns C1–C8 and the normalization conditions, steps 10–14 by campaigns C9–C12 and the Evidence Engine, step 15 by the reproducibility pipeline and the audit. No step carries a partially-done status: where the result is negative (steps 5 and 6), the negative is recorded as a full-fledged result with a falsifier.

3. Step 10 ↔ C9: the phase map

The tenth step demanded a systematic scan — not spot checks — of the question whether the gap closes anywhere. Campaign C9 discharges it with a 9×41 = 369-cell map: all OPEN. The statement is deliberately finite-range — it concerns the explored window beta ∈ [2.80, 4.55], L ∈ [6, 32] — and precisely therefore does not become an uncontrolled generalization.

The second part of the step is protection against extrapolation self-deception. The exploratory build's naive window predicted closure at beta_c = 2.9910; the direct probe measured 6.3 thresholds there (claim C9-FALS, honest-negative). The lesson is embedded in the documentation and in the tests: any future extrapolator must pass a direct probe check. Artifacts: c9_phase_grid.json, c9_valley.json, c9_naive_falsifier.json; tests test_c9_c10.py; figures fig_phase_map_*.

4. Step 11 ↔ C10: competing models

The eleventh step required not picking the best model but honestly weighing the competing causal explanations of the signal. C10 decomposes the signal into three channels and measures the shares: A = 91.95 % (propagating), B = 8.00 % (braking), C = 0.049 % (contact). The additivity bias 1.3e-4 is separated from the shares and documented; the braking estimator B2 vanishes on the even control exactly — a SymPy identity, not a numerical accident.

The negative matters equally: out of window the model has no validity, and the residuals grow as (t−t1)^4.071. This result prevents the typical mistake of extrapolating a fit beyond its training window. The step's falsifier: the emergence of a stable fourth channel above 0.01 %, or a residual power outside [3.5, 4.7]. Artifacts: c10_shares.json, c10_braking.json, c10_out_of_window.json.

5. Step 12 ↔ C11: the classical anchor

The twelfth step required a classical layer whose stability is provable rather than declared. C11 provides three provable elements: the spectrum {0, 0, ±iomega, ±iomega} with its symbolic polynomial (T11, exact), the Aref boundedness with the exact bound 2·sqrt(I) and the measured r_max = 2.7654 ⋘ 12 (T12), and the static barrier V* = 0.9359375 with the maximum exactly at s = 1 and zero escapes out of three hundred trajectories (T13).

The anchor plays the role of the classical conservative in the program: if even the simplest classical dynamics already holds the configuration spectrally and barrier-wise, the only quantum escape channels are tunneling ones — and C7 closes those. Falsifiers: Re lambda ≠ 0, a single escape with E < V*, an invariant drift above 1e-12. Artifacts: c11_spectrum.json, c11_rotation.json, c11_aref_run.json, c11_escape_scan.json.

6. Step 13 ↔ the Evidence Engine

The thirteenth step is organizational: convert all results into a single registry of falsifiable claims. The Evidence Engine holds thirty claims in five statuses; each carries a falsifier and a five-field uncertainty budget (statistical, systematic, model, numerical, external). Hypotheses (HYPO-CONT) are stored separately and never cited as results; external values (eta_B, nugget bounds) carry the comparison status only.

The registry is the interface between computation and reading: the documentation, the monographs, the portal and the workbook reference claims by stable IDs. Changing a number in a campaign without updating the registry breaks the tests — artifact consistency is machine-checked. Artifacts: claims.json, audit_ivp12.json, twins.json, rebuild_hashes.json.

7. Step 14 ↔ C12: the asymmetry lock

The fourteenth step is the substantive summit of the program: prove that the gap is permanent and free quarks are impossible. C12 discharges it in three moves. Move one (T14, exact): for any asymmetry eta < eta_crit = Delta_F,min/2 = 0.0908885 the closure barrier stays positive; the worst profile loses exactly 2eta; the cosmological value eta_B = 6.1e-10 sits 1.49e8 times below the threshold. The 41-point eta map confirms the boundary numerically.

Move two (T15, honest-negative): the observational route of free quarks is closed — gap filling self-limits through negative feedback, zero saturated wells in 10^6 steps at eta_B; the positive control at eta = 0.5 cascades, demonstrating that the mechanism is not hard-coded. Move three (comparison): the combinatorial probability of a macroscopic accumulation is below 10^(−1.2e31), and the experimental search bounds agree with the zero prediction.

The step's summary: because of the complete matter–antimatter asymmetry the gap will always exist, and quarks cannot exist separately — scattering in the quantum field and filling the gaps, they would produce isolated large accumulations which are not observed; within the QHB model class this argument is proven quantitatively, with falsifiers, and without leaving the declared domain of applicability.

8. Step 15 and the open items

The fifteenth step is the release step: a reproducible package, an audit, bilingual mirrors, tests. It is discharged by the make all pipeline: 31 tables with byte-identical rebuilds, IVP-12 12/12, 51 twin checks, 154 tests, determinism without drift. The release ships with monographs, the workbook and this document; all artifacts reference the claims registry.

The open items are listed honestly: the continuum limit remains a hypothesis (step 2 is finite-volume); the detector tail shape is unresolved; unequal vortex circulations and the temperature dependence of the lock are roadmap candidates R2–R3; the external comparison with the nugget literature needs proper references (R5). These items do not affect the discharged steps 10–14: their falsifiers do not intersect.

Table 8.1 — Open items and their falsifiers

Item	Type	What would close it
Continuum gap Delta_cont	hypothesis	L-convergence beyond 32
Tail shape beyond E_max	honest-negative	Detector resolution (new C4 model)
Unequal circulations	roadmap R2	Splitting of the ±iomega pair

Item	Type	What would close it
Temperature lock eta(T)	roadmap R3	Activation decay of the barrier
Nugget literature	roadmap R5	Formatted references for C12-EXP

The falsifier-first policy extends to the roadmap: every new item starts with the statement of what would refute it.